Q1: Check whether the following two fields are Quasi-static or rapidly time varying field. (20 mark)

1- Distance between transmitter and receiver is 10km and the frequency is 30k signal carrier.

2- The time lag is 17ps and the frequency is 3GHz

Q2: List the modern technology applications of electromagnetic waves (only 5) (20 mark)

Q3: Specify the unit vector extending from the origin toward point G (2, -2, -1) (20mark)

Q4: Electric current density (J measured in A/m2) can be seen in different forms, list Them (Electric Current Types). (20 mark)

Q5: Write Maxwell's equations in point form in terms of E and H as they apply to a source: medium, where J and B are both zero. Replace € by M, E by H, and H by d. show that the equation unchanged. (20 mark)

Q6: Discuss the fundamental differences between static and dynamic electromagnetic fields. Explain how their time behavior, source mechanisms, and field interactions influence their ability to support wave propagation and their suitability for communication systems. (20 mark)

Q7: Field Classification (20 mark)

1- system operates at 100 kHz with size 500 m:

(A) Find wavelength. (B) Classify the field. (C) Explain physically.

2- Time Analysis Given: f=4"GHz",τ=40"ps"

(A) Find T (B) Classify (C) Effect of doubling delay

Q8: Write Maxwell's equations in point form in terms of E and H as they apply to a source: medium, where J and B are both zero. Replace € by M, E by H, and H by d. show that the equation unchanged. (20 mark)

Q9: Answer five questions below (MCQ) (20 Mark)

In ferromagnetic materials, the relative permeability (μr) is:

1. Less than 1 2. Much greater than 1 3. Equal to 1 4. Equal to zero

B. The magnetic force on a moving charge (Q) with velocity (u) in a magnetic field (B) is given by:

1. F = Q / B 2. F = Q (u. B) 3. F = Q (u × B) 4. F = u (Q × B)

C. According to magnetic boundary conditions, the normal component of magnetic flux density(B) is: 1. Discontinuous 2. Always zero 3. Continuous across the boundary. 4. Equal to the tangential component.

D. The Lorentz Force Equation combining both electric and magnetic forces is:

1. F = Q (E + u × B) 2. F = Q (E- u × B) 3. F = Q (E × B) 4. F = Q (E. B) + u

According to Faraday's law, a time-varying magnetic field generates: E.

1. Direct current 2. An electric field (∇ × E =-∂B/∂t) 3. Static charges. 4. Another magnetic field

G. Gauss's law for magnetism (∇. B = 0) physically implies:

1. Isolated magnetic poles exist 2. Non-existence of magnetic monopoles 3. Magnetic field is constant

4. Charge is conserved.

Q10: List and briefly define the main types of electric current density.(20 mark)

Q11: List the modern technology applications of electromagnetic waves.(20 mark)

Q12: compare between the electric force and the magnetic force? (20 mark)

Q13: Answer five questions below (MCQ) (20 mark)

A. In ferromagnetic materials, the relative permeability (μr) is:

1. Less than 1 2. Much greater than 1 3. Equal to 1 4. Equal to zero

B. The magnetic force on a moving charge (Q) with velocity (u) in a magnetic field (B) is given by

1. F = Q / B 2. F = Q (u. B) 3. F = Q (u × B) 4. F = u (Q × B)

C. According to magnetic boundary conditions, the normal component of magnetic flux density B is: 1. Discontinuous. 2. Always zero. 3. Continuous across the boundary. 4. Equal to the tangential component.

D. The Lorentz Force Equation combining both electric and magnetic forces is:

1. F = Q (E + u × B) 2. F = Q (E- u × B) 3. F = Q (E × B) 4. F = Q (E. B) + u

According to Faraday's law, a time-varying magnetic field generates: E.

1. Direct current 2. An electric field ($\nabla \times E = -\partial B/\partial t$) 3. Static charge 4. Another magnetic field .

G. Gauss's law for magnetism ($\nabla . B = 0$) physically implies:

1. Isolated magnetic poles exist. 2. Non-existence of magnetic monopoles. 3. Magnetic field is constant 4. Charge is conserved.

Q14: List and briefly define the main types of electric current density. (20 mark)

Q15: List the modern technology applications of electromagnetic waves. (20 mark)

Q16: Discuss the fundamental differences between static and dynamic electromagnetic fields. Explain how their time behavior, source mechanisms, and field interactions influence their ability to support wave propagation and their suitability for communication systems.? (20 mark)

ANS/ **1/**

Maxwell Equations:

$\nabla \cdot \mathbf{D} = \rho$ Gauss Law for electric field.

$\nabla \cdot \mathbf{B} = 0$ Gauss Law for magnetic field.

$\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$ Faraday's Law.

$\nabla \times \mathbf{H} = \mathbf{J} + \partial \mathbf{D} / \partial t$ Ampere's Law.

Where:

D: electric displacement (c/m^2), E: electric field (V/m),

B: magnetic field (T or $V.s/m^2$),

ρ: Volume charge density (c/m^3),

2/ A-2 B-3 C-3 D-1 E-2 G-2

3/

$$T = \frac{1}{4 \times 10^9} = 250 \text{ ps}$$

الشرط غير محقق
Dynamic Field

$$\lambda = \frac{3 \times 10^8}{10^5} = 3000 \text{ m}$$

Dynamic Field ✅

(D) التفسير الفيزيائي

- الأبعاد ليست صغيرة مقارنة بالطول الموجي
- يظهر تغير في الطور
- تبدأ تأثيرات الموجة بالظهور

(D) عند مضاعفة التأخير

$\tau = 80 \text{ ps}$

Dynamic Field ✅

4/

.Impressed current density or called the source current density (JS) -1 /4
-Convection current density (Jv) -2
Induced current density, which is the density of the current that is -3 -3 caused by
a- Conduction current density (Jc), which is the density of the current that is
b- Displacement current density (Jd), c- Polarization ,current density (Jp)

5/ Wireless Communication (Wi-Fi, 5G, Phones): Information is transmitted through the air as electromagnetic waves. Antennas convert electrical signals into waves and vice-versa, based on the interrelation of electric and magnetic fields. Wireless Charging:Uses Induction. A charging pad creates a changing magnetic field that induces an electric current in your phone's coil to charge the batter

Microwave Ovens: The oscillating electric field of the microwaves forces water molecules in food to rotate rapidly, generating heat through friction to cook the food. Electric Motors & Generators: Generators convert mechanical motion into electricity (Faraday's Law), while motors convert electricity into motion using magnetic forces. MRI Machines :

/ The electric force - 6/ is independent of the velocity of the charg

• The electric force 2 -

can perform work on the charge and change its kinetic energy .

• The magnetic force 3

depends on the charge velocity and is normal to it .

• The magnetic force

cannot perform work because it is at right angles to the direction

of motion of the charge 4

;) it does not cause an increase in kinetic energy of

6/

Solution:

1- Here the distance is given therefore it is more convenient to use equation (2) where we compare D_{max} with λ. D_{max} is given as 10km = 10^4m. The frequency is 30kHz = $3x10^4$ so the wavelength can be found as

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{3 \times 10^4} = 10km$$

This means that D_{max} is NOT much smaller than the wavelength. Hence, equation (2) does NOT apply, which means that this field is a high-frequency dynamic field.

2- The time is given; therefore, it is easier to use equation (1) where $\tau = 17x10^{-12}$ s and frequency is $3x10^9$ Hz. The period is

$$T = \frac{1}{f} = \frac{1}{3 \times 10^9} = 333.3\ ps$$

Here equation (1) is verified since the period is much greater than the time lag. This means that this is a low-frequency (Quasi-static) field.

Notice that this example has shown that the low- and high- frequency fields do not mean that the source frequency is low or high. The frequency should be compared according to any of equations (1) and (1).

7/

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8/

$$\mathbf{G} = 2\mathbf{a}_x - 2\mathbf{a}_y - \mathbf{a}_z$$

We continue by finding the magnitude of **G**,

$$|\mathbf{G}| = \sqrt{(2)^2 + (-2)^2 + (-1)^2} = 3$$

and finally expressing the desired unit vector as the quotient,

$$\mathbf{a}_G = \frac{\mathbf{G}}{|\mathbf{G}|} = \tfrac{2}{3}\mathbf{a}_x - \tfrac{2}{3}\mathbf{a}_y - \tfrac{1}{3}\mathbf{a}_z = 0.667\mathbf{a}_x - 0.667\mathbf{a}_y - 0.333\mathbf{a}_z$$

.Impressed current density or called the source current density (JS) -1 /4
-Convection current density (Jv) -2
Induced current density, which is the density of the current that is -3 -3
caused by
a- Conduction current density (Jc), which is the density of the current
that is
b- Displacement current density (Jd), c- Polarization
,current density (Jp)

9/ 1. Definition and Time Dependency

Static electromagnetic fields are time-invariant, meaning that their magnitude and direction do not change with time. Consequently, they are associated with zero frequency (DC fields) .

In contrast, dynamic electromagnetic fields are time-dependent, where field quantities vary with time and are associated with non-zero (often high) frequencies.

2 Wave Propagation ..

Static fields have negligible radiation and therefore do not support wave propagation. As a result, they cannot transmit energy over long distances in the form

of waves . On the other hand, dynamic fields support propagating electromagnetic waves, which enables energy and information transfer through space

3 . Field Coupling

In static conditions, the electric field and magnetic field are not coupled, meaning each can exist independently and be analyzed separately .

However, in dynamic fields, the electric and magnetic fields are coupled through the mechanism of mutual induction. A time-varying electric field generates a magnetic field and vice versa, making them interdependent and inseparable in analysis .

4 . Sources of Fields

Static electric fields (electrostatic fields) are generated by stationary charges, while static magnetic fields (magnetostatic fields) are produced by steady (constant) electric currents .

In contrast, dynamic electromagnetic fields are generated by time-varying charges and currents, which lead to continuously changing electric and magnetic fields .

5 . Communication Applications

Due to the absence of wave propagation, static fields cannot be used in communication systems .

Conversely, dynamic electromagnetic fields form the basis of modern communication systems, as they propagate as waves capable of carrying signals over long distances (e.g., radio waves, microwaves)